

Nexus Carrier Hotel (MMR) and Data Center Connectivity Design

1. Introduction

This document provides a high-level design overview of the Nexus carrier hotel / Meet-Me Room (MMR) architecture and the connectivity model into Nexus data center buildings.

It describes how carrier fiber enters the environment, how cross-connects are established, and how connectivity is extended into Nexus buildings using MaiaEdge Path Border Controllers (PBC).

This document is intended for data center engineering, planning, and financial review.

2. Design Goals

- Provide a repeatable model for multi-carrier environments.
- Support initial deployment across 4 buildings, scalable to 18 buildings.
- Support up to 18 carriers.
- Enable high availability via diverse fiber paths and dual-device architecture.
- Align with real-world carrier hotel and data center practices.

3. High-Level Architecture

The architecture consists of two primary domains:

- **Carrier Hotel / MMR** — Interconnection Domain
- **Nexus Building / Data Center** — Consumption Domain

4. Carrier Hotel / MMR Design

The MMR acts as the central interconnection hub where carrier fiber is terminated and cross-connected.

Key components:

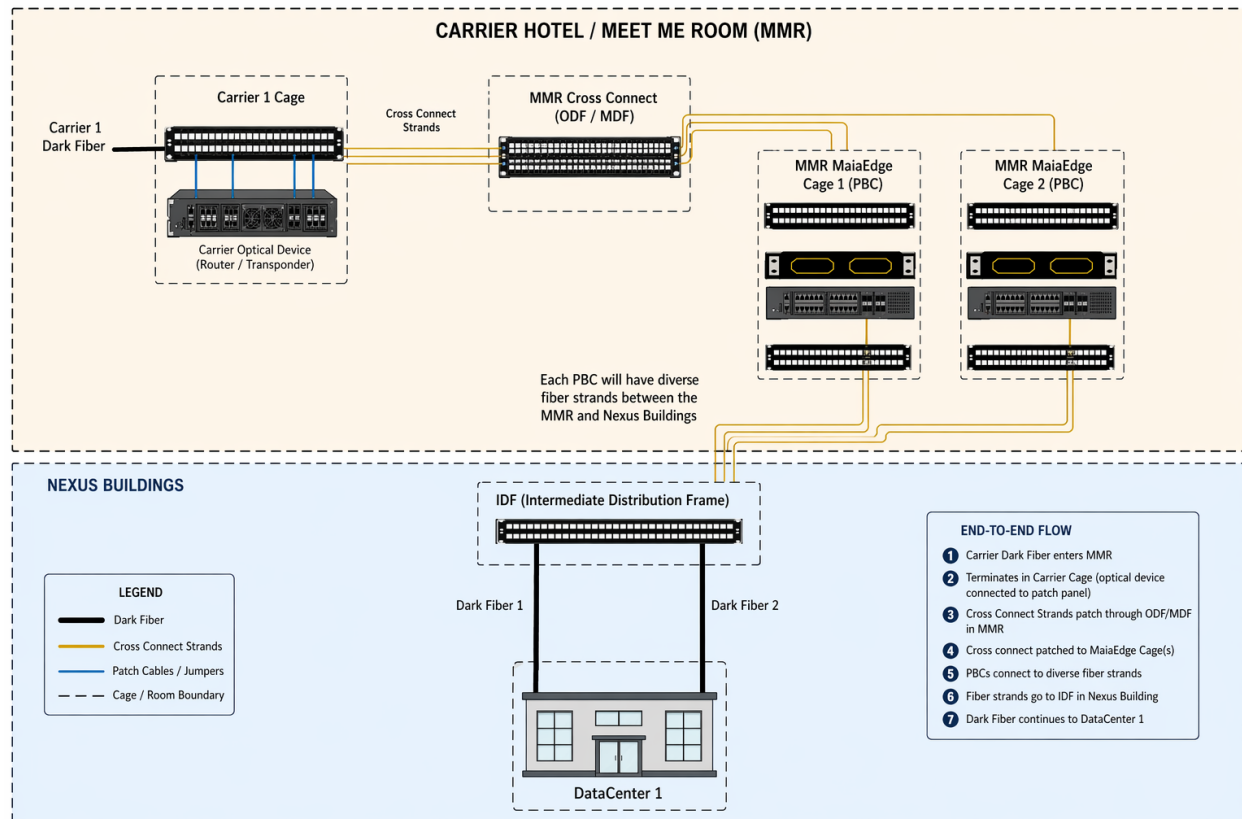
- Carrier fiber entry into the facility
- Carrier cages with optical devices and patch panels
- MMR Optical Distribution Frame (ODF) for cross-connects
- MaiaEdge cages containing PBC devices

Signal Flow

The signal flow starts from the carrier fiber coming into the building.

Carrier Fiber Ingress

- Carrier Optical Device (Router / Transponder)
- Carrier Patch Panel
- MMR Cross-Connect (ODF)
- MaiaEdge Patch Panel
- **Path Border Controller (PBC)**



MMR Network Diagram

Stage	Component	Description
1	Carrier Fiber	Fiber enters facility from carrier network
2	Optical Device	Carrier router/transponder generates signal
3	Patch Panel (Carrier)	Local termination within carrier cage
4	ODF (MMR Cross-Connect)	Central fiber cross-connect fabric
5	Patch Panel (MaiaEdge)	Termination inside MaiaEdge cage
6	PBC	Programmable edge for path control

5. Nexus Data Center / Building Design

Connectivity is extended from the MMR into Nexus buildings via diverse dark fiber paths.

Key components:

- Diverse dark fiber paths (A/B) from MMR
- IDF / ZDA for passive fiber distribution
- MaiaEdge PBC cages (dual for redundancy)
- Leaf/Spine switch fabric
- ToR switches and client workloads

Signal Flow

Signal flow for the datacenter starts from the dark fiber.

Dark Fiber (from MMR)

→ IDF / ZDA (Fiber Distribution – Passive)

→ Patch Panel

→ **MaiaEdge PBC (Active Edge)**

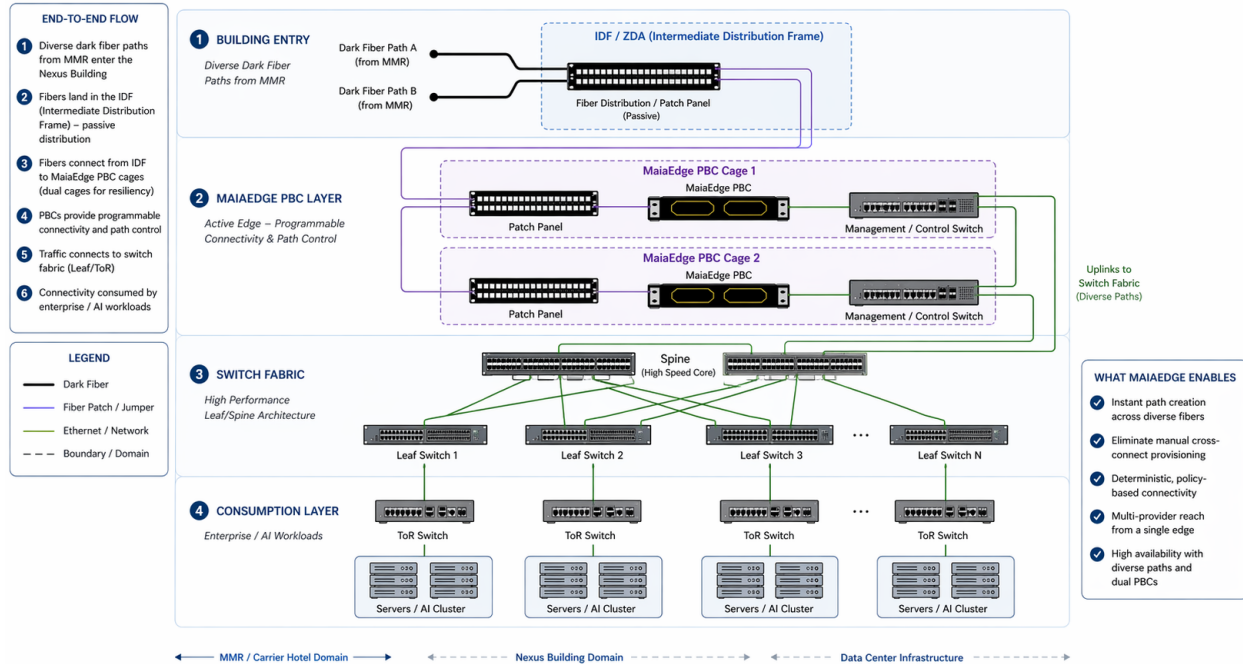
→ Leaf / Spine Switch Fabric

→ Top-of-Rack (ToR) Switch

→ **Servers / AI / Enterprise Workloads**

NEXUS DATACENTER – CONNECTIVITY ARCHITECTURE

Dark Fiber from MMR → IDF → MaiaEdge PBC Cages → Switch Fabric → Clients



Datacenter Building Network Diagram

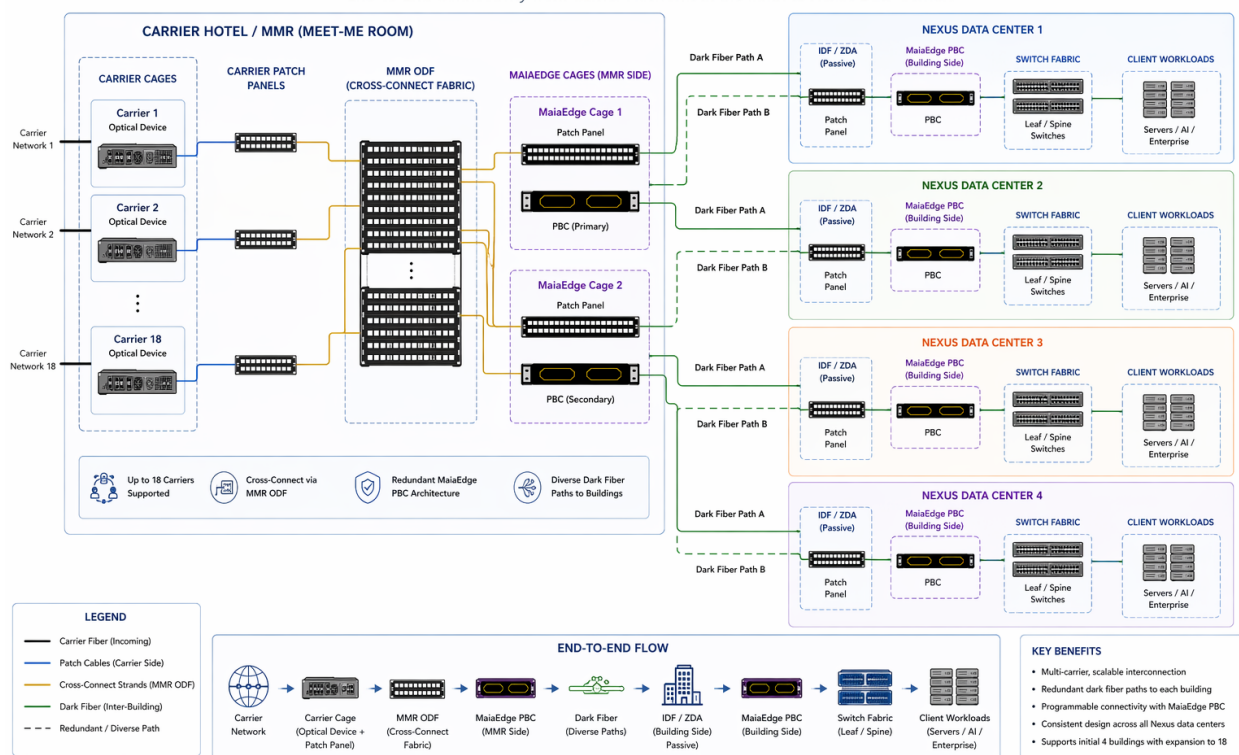
Stage	Component	Description
1	Dark Fiber	Diverse fiber paths from MMR into building
2	IDF / ZDA	Passive fiber distribution and termination
3	Patch Panel	Local fiber termination and cross-connect
4	PBC	Programmable edge for path control
5	Leaf / Spine	High-speed switching fabric
6	ToR Switch	Rack-level aggregation
7	Servers	AI / enterprise workload consumption

6. End to End Diagram

Connectivity is extended from the MMR into multiple Nexus buildings via diverse dark fiber paths.

MMR CONNECTED TO FOUR NEXUS DATA CENTERS OVER DIVERSE DARK FIBER

End-to-End Connectivity from Carrier Networks in the MMR to Nexus Data Centers



Multi-Building Nexus Connectivity via MMR over Diverse Dark Fiber

6. Redundancy Model

- Dual PBC deployment for high availability
- Diverse fiber paths (A/B)
- Independent physical routing to eliminate single points of failure

7. Key Differentiation

Traditional Model

- Manual cross-connect provisioning
- Static connectivity

Nexus + MaiaEdge Model

- Programmable connectivity
- Instant path creation
- Deterministic routing
- Multi-provider reach